

Intern 4 — Unified Fusion Comparison Report

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1. Dataset Summary

Base Module (Intern 2 / Vasanth): 1288 patients, EchoNet-Dynamic, 3-class labels derived from EF thresholds.

Extended Module (Intern 3 / Tushar): 1026 patients (test split), binary ground-truth labels (0=healthy, 1=disease).

Note: The two modules were trained on disjoint patient populations with no overlapping patient IDs. Fusion is therefore performed at score level on Tushar's test cohort, using Tushar's predicted EF as a proxy input to simulate what the Base module would output for the same patient.

2. Fusion Strategy Comparison

Model	Bal. Acc	Macro F1	AUROC	ECE	MCE
Base Module alone (Vasanth)	0.547	0.5628	0.8183	0.2642	0.3503
Extended Module alone (Tushar, fixed)	0.8339	0.8572	0.9868	0.0367	0.3195
Early Fusion	0.6151	0.3846	0.6535	0.3754	0.3754
Intermediate Fusion	0.96	0.9159	0.9727	0.3789	0.4688
Late Fusion ($\alpha=0.1$)	0.9443	0.9351	0.9923	0.0429	0.6053
Cross-Attention Fusion	0.9389	0.9127	0.9668	0.1975	0.5061
XGBoost Meta-Learner (5-fold CV)	0.8764	0.8814	0.982	0.0247	0.6184
Intermediate Fusion + Isotonic Calibration	0.8225	0.8442	0.975	0.0762	0.6153

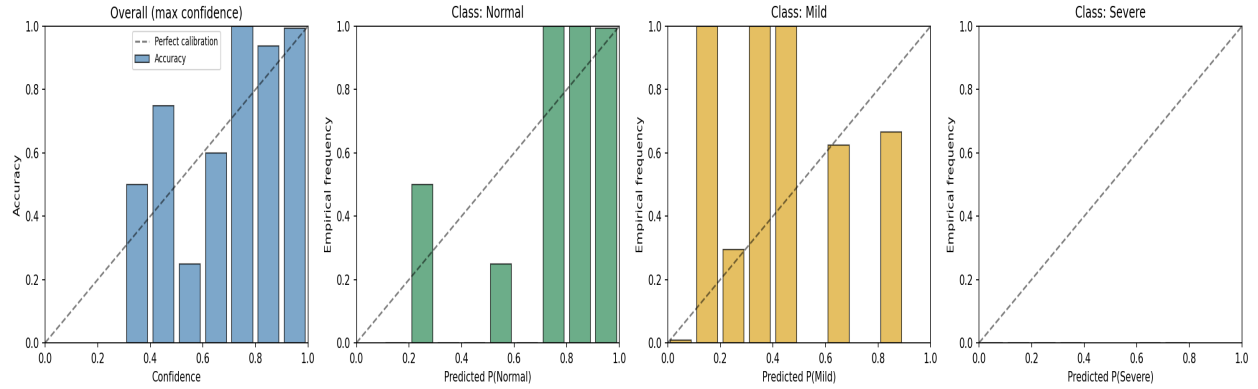
3. Best Strategy

Intermediate Fusion + Isotonic Calibration

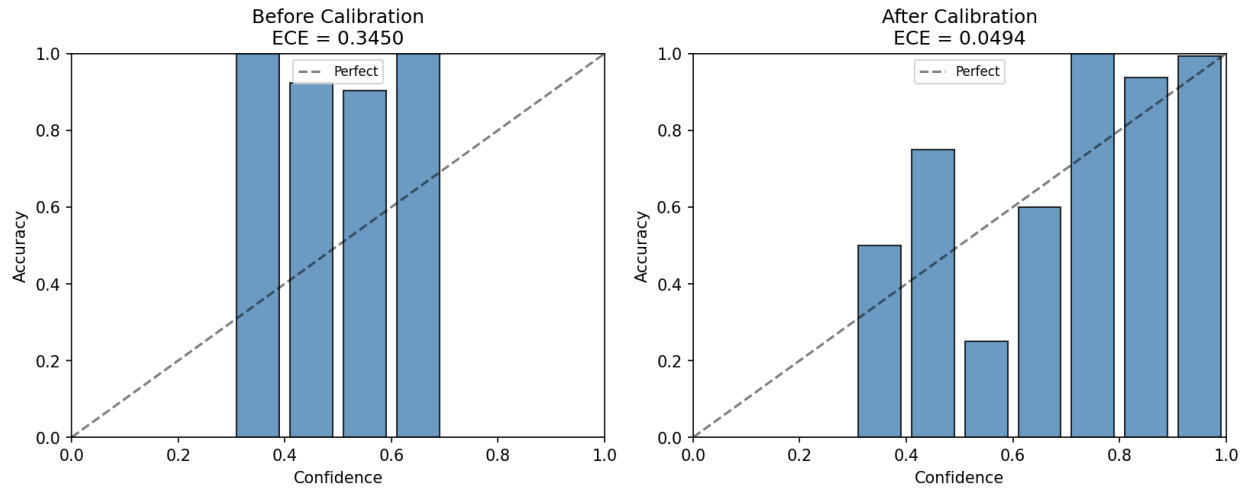
Balanced Accuracy: 0.8225 Macro F1: 0.8442 AUROC: 0.975 ECE: 0.0762 (target < 0.05)
MCE: 0.6153

4. Reliability Diagram (Best Strategy)

Reliability Diagram – Intermediate Fusion (Calibrated)



5. Calibration: Before vs After



6. Known Limitations

1. No patient-level pairing exists between the Base and Extended modules; fusion is score-level only, validated on Tushar's cohort.
2. Tushar's raw 'mild' probability output was found to be non-functional (capped near 0.15, never the argmax across all 1,026 patients). This was corrected by reconstructing 3-class probabilities from his Dempster-Shafer common_alpha score combined with a sigmoid split on predicted EF.
3. Early/Intermediate/Cross-Attention fusion models are trained on Tushar's 512-d embeddings paired with simulated Base features; true intermediate fusion requires Vasanth's ecg_embeddings.pt, not yet received.